SECOND-HAND VEHICLE SELLING WEBSITE

Learning Block 1

Submitted by

Mohammad Arshad

Kommuri Pratap Reddy Institute of Technology

Department of Computer Science Engineering

Academic Year: [Final] – [2026–2027]

Guided by: Raghavendra Rao M S

Index

Index	Page No.
1. Introduction	1
2. Problem Statement	2
3. Objectives	3
4. Project Scope	4
5. Proposed System / Methodology	5
6. System Architecture / Workflow	6
7. Implementation	7
8. User Interface / Application Screenshots	8
9. Challenges and Limitations	11
10. Conclusion	12
11. Future Scope	13
12. References	14
13. GitHub Repository Link	15

1. Introduction:

The Second-Hand Vehicle Selling Website is a full-stack web application designed to make buying and selling pre-owned vehicles more convenient. The platform provides a common online space where users can browse listed vehicles, review important details, and contact sellers. Sellers can submit vehicle information so that potential buyers can discover suitable options more easily.

Buying a used vehicle can involve searching across multiple websites, classified advertisements, and local dealers. Details such as price, model year, kilometres driven, fuel type, transmission, vehicle condition, and location may be presented inconsistently. This can make it difficult for buyers to compare available options and identify vehicles that match their needs.

The proposed website organizes vehicle listings in a structured format and offers search and filtering features. It covers a mix of second-hand cars, bikes, and trucks. The frontend is built with HTML, CSS, and JavaScript, the backend uses a Node.js server implemented in server.js, and MySQL is used for persistent data storage.

The project demonstrates the integration of client-side interface design, server-side request handling, and relational database operations in a practical full-stack development application.

2. Problem Statement:

The traditional process of buying or selling a second-hand vehicle can be time-consuming. Buyers may need to visit several dealers or browse different online listings to compare prices, vehicle condition, mileage, and location. Sellers may also find it difficult to reach interested buyers and present complete vehicle information in one place.

Unstructured advertisements can make comparisons difficult and may omit important specifications. A centralized website can address these issues by presenting vehicle details consistently and allowing users to search and filter listings according to their preferences.

The project therefore aims to develop a web-based platform where vehicle information can be displayed and managed through a frontend, backend server, and MySQL database. The website is intended to improve convenience and organization; users should still verify vehicle documents, condition, ownership, and seller details independently before completing a purchase.

3. Objectives:

- To develop a user-friendly full-stack website for listing, browsing, and comparing second-hand vehicles.
- To create a simple and responsive interface for browsing second-hand vehicles.
- To allow users to search listings by vehicle name, brand, model, and location.
- To provide vehicle specifications such as price, model year, kilometres driven, fuel type, and transmission.
- To support vehicle listing submission through a seller form.
- To connect the frontend to a Node.js backend using HTTP/API requests.
- To use MySQL for storing and retrieving vehicle listing records.
- To provide clear feedback for successful requests, invalid input, and server or database errors.
- To provide favorites, vehicle comparison, and user login features for a better user experience.

4. Project Scope:

The project focuses on a web application for second-hand vehicle listings. The project covers the following activities:

- Displaying vehicle listings for second-hand cars, bikes, and trucks.
- Viewing individual vehicle details and seller-provided information.
- Searching and filtering listings by fields such as brand, price, type, year, fuel, transmission, and location.
- Submitting vehicle information through a seller listing form.
- Using server.js to receive requests and coordinate application logic.
- Connecting the backend to MySQL for storing and retrieving vehicle listing records.
- Providing a responsive layout suitable for common desktop and mobile screen sizes.

5. Proposed System / Methodology:

The proposed system uses a three-layer architecture: frontend, backend, and database. The frontend provides the pages and forms through which users interact with the application. JavaScript handles interface interactions and sends requests to the backend. The Node.js server in server.js receives requests, validates input, executes the relevant application logic, and communicates with MySQL.

When vehicle information is requested, the backend queries the database and returns the result to the frontend. When a seller submits a listing, the server validates the submitted fields and can store the vehicle record in MySQL. The frontend then displays appropriate success or error feedback.

End-to-End Process

1. **User Interaction:** A buyer browses vehicles or a seller enters vehicle details.

2. **Frontend Processing:** HTML structures the page, CSS controls presentation, and JavaScript validates inputs and sends requests.

3. **Backend Request:** server.js receives the request through the appropriate route.

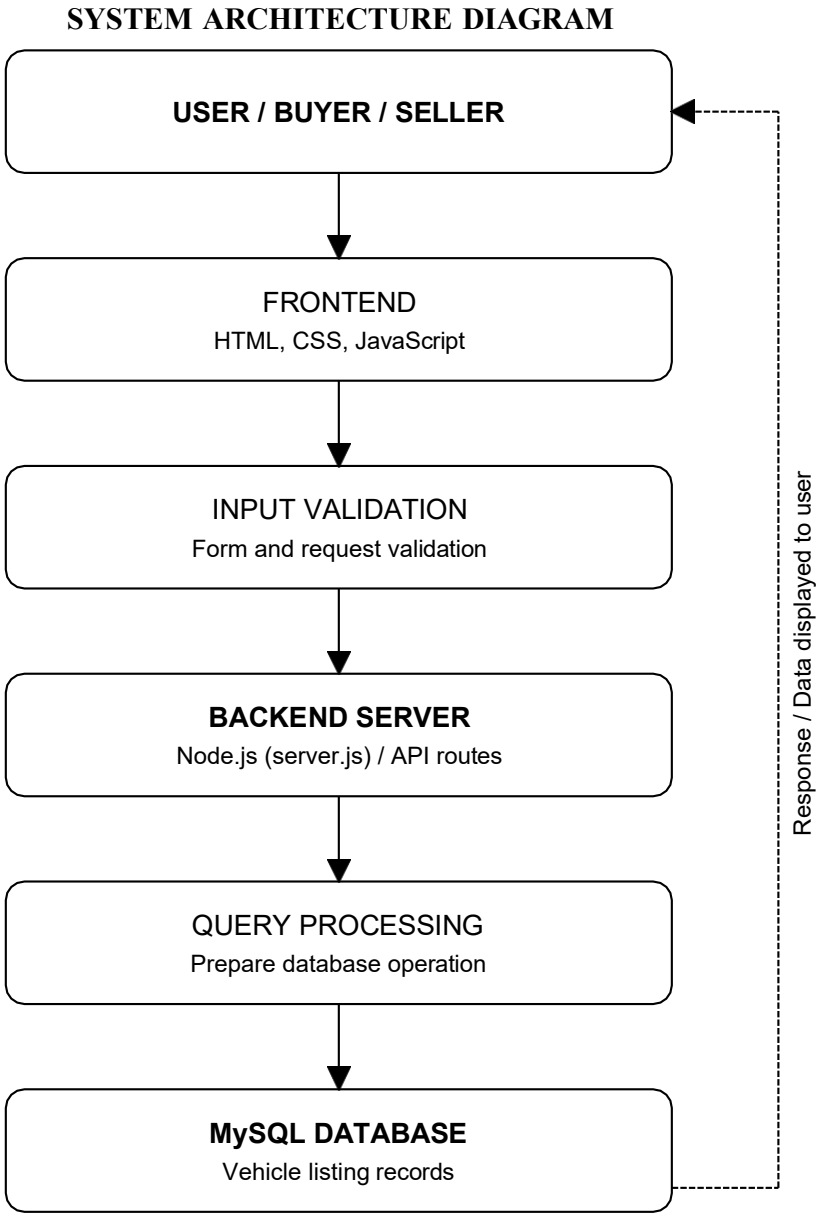
4. **Validation and Processing:** The backend checks required fields and prepares the database operation.

5. **Database Operation:** MySQL stores, updates, or retrieves records as required.

6. **Response:** The server returns a success message or the requested data, or reports an error.

7. **Display:** The frontend updates the page so the user can view listings or submission status.

6. System Architecture / Workflow:



Architecture Explanation:

The user searches for vehicles or submits a form through the frontend. The frontend sends the request to the Node.js backend (server.js), which validates it and performs the required MySQL operation. The database returns the result, and the backend sends a response that the frontend displays.

7. Implementation:

The website is developed using the following full-stack technologies.

Software Components

- **Frontend (HTML):** defines the structure of pages, forms, navigation, and vehicle listing cards.
- **Frontend (CSS):** provides layout, colors, typography, spacing, responsive behavior, and visual styling.
- **Frontend (JavaScript):** manages interactive controls, basic form validation, search/filter interactions, and communication with backend endpoints.
- **Backend (Node.js, server.js):** runs the server, defines API routes, handles JSON requests, applies validation, and coordinates database operations.
- **Database (MySQL):** stores vehicle details in structured tables and supports retrieval and maintenance of records.
- **Testing:** the frontend, backend, and database are tested together to confirm that requests, responses, and stored data work as expected.

Functional Workflow

- The user opens the website and browses or submits vehicle details.
- The frontend sends a request to the backend.
- server.js processes the request and validates the input.
- MySQL is queried or updated.
- The response is displayed in the interface.

Security considerations include keeping database credentials out of screenshots and source-code examples, using parameterized SQL queries, and handling invalid input and database errors gracefully.

8. User Interface / Application Screenshots:

The following screenshots show the implemented Auto Bazaar second-hand vehicle website running in the browser.

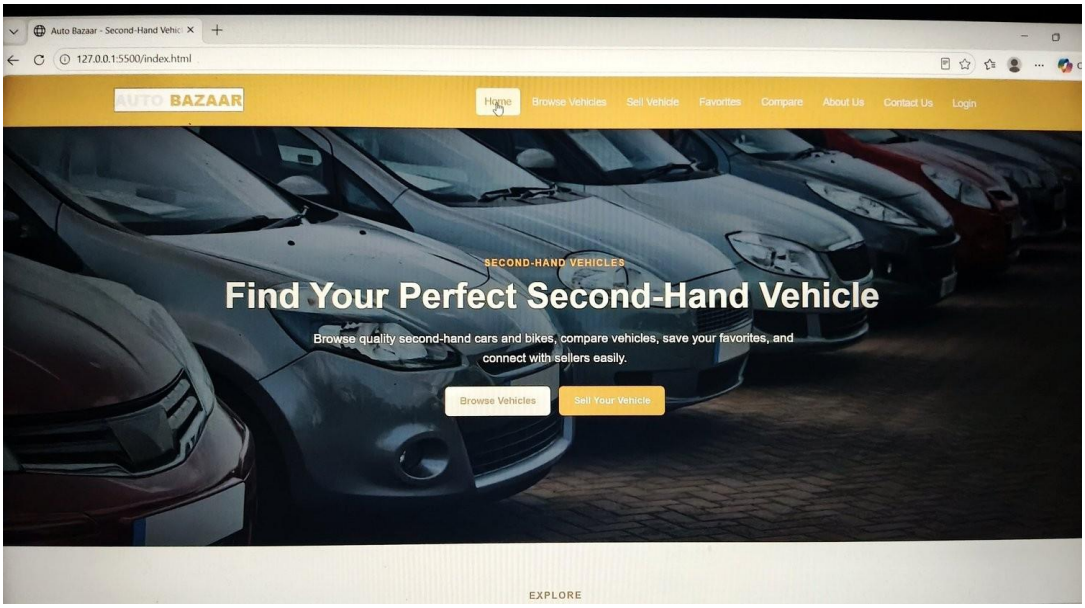


Figure 1: Home page

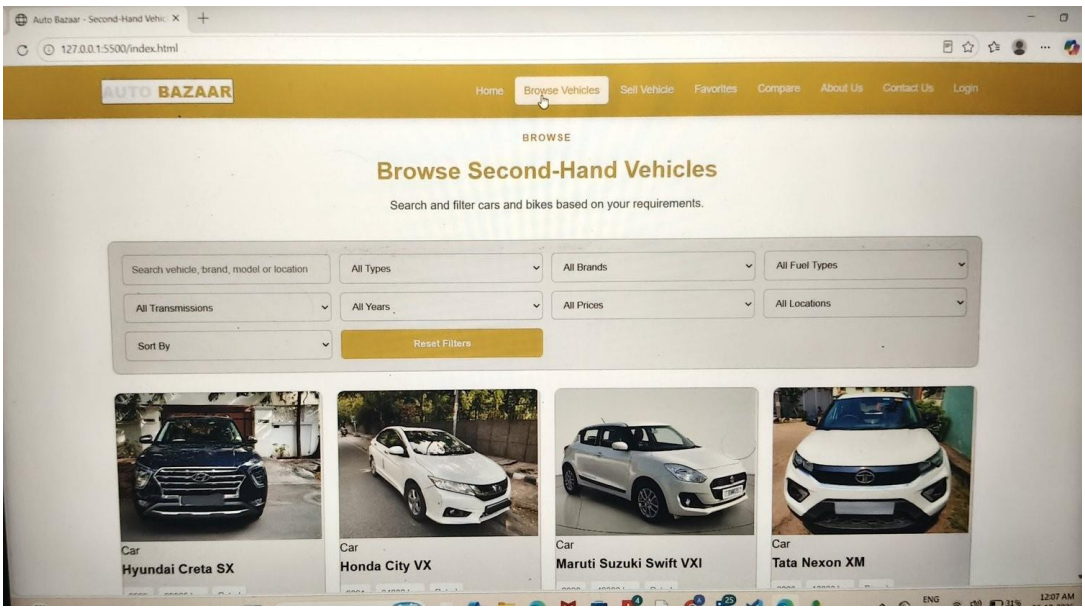


Figure 2: Browse Vehicles page with search and filter controls

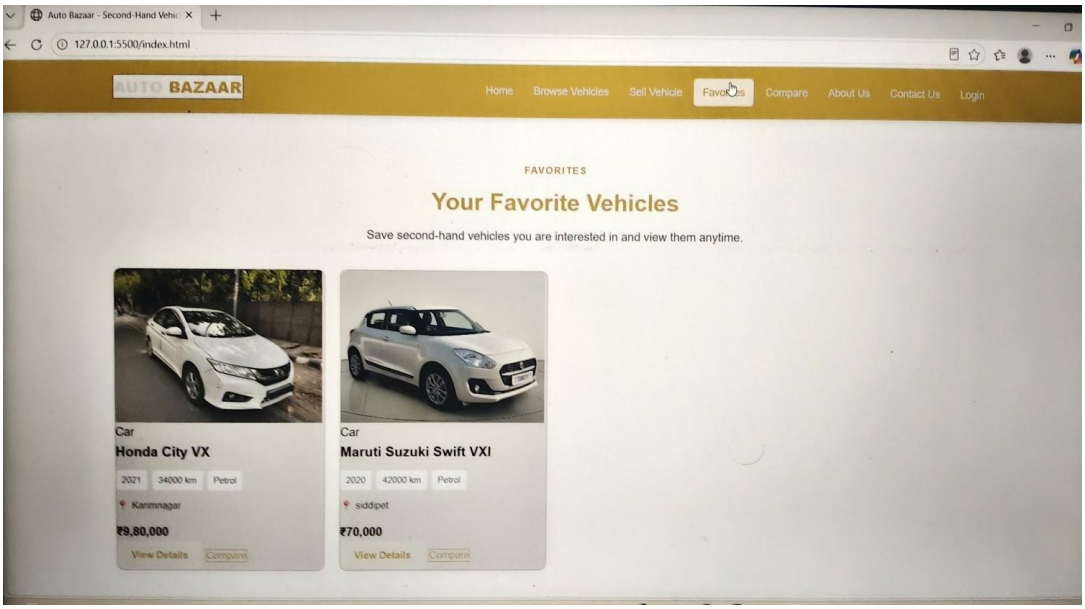


Figure 3: Favorites page showing saved vehicles

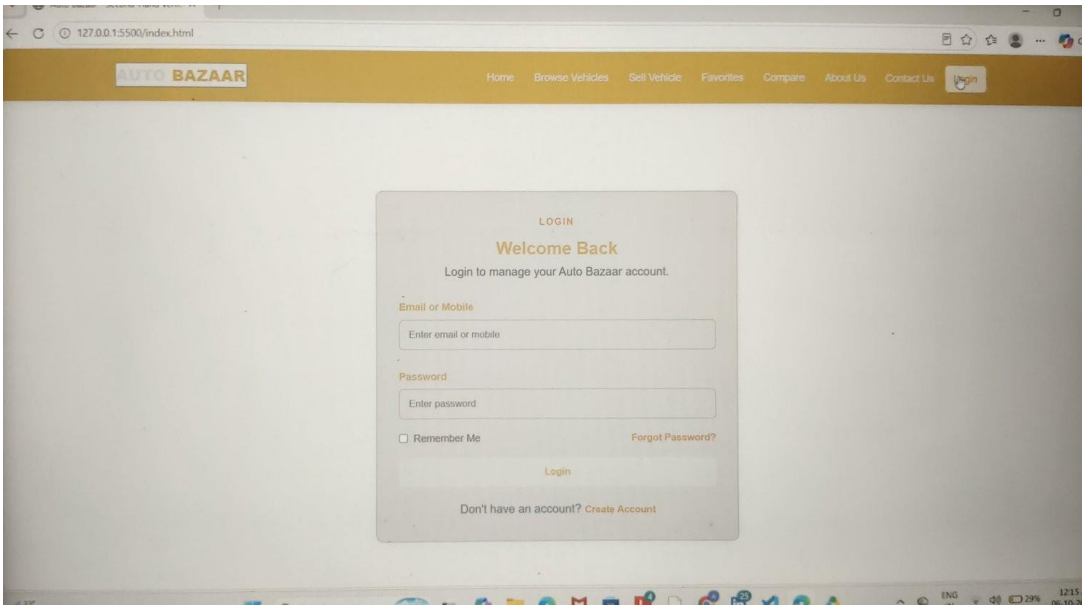


Figure 4: Login page

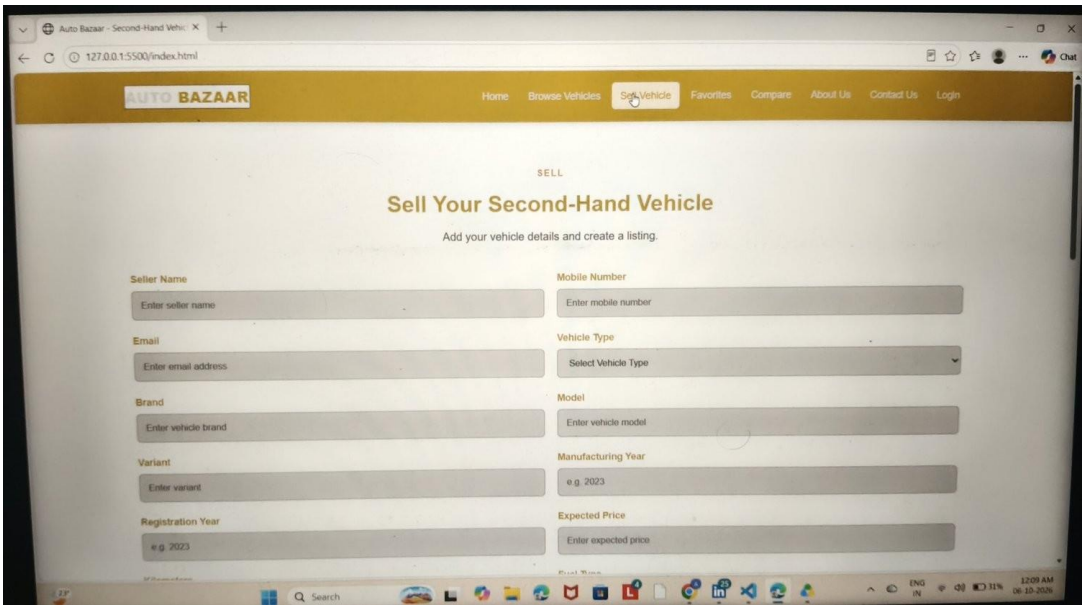


Figure 5: Sell Vehicle page with the seller listing form

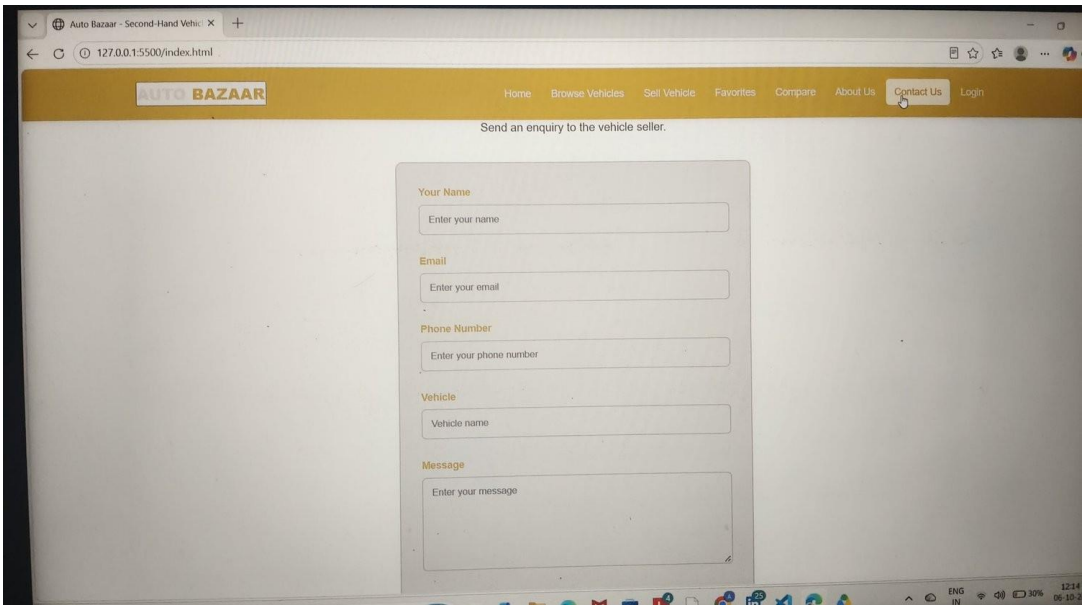


Figure 6: Contact Us page with the seller enquiry form

9. Challenges and Limitations:

Challenges

- Keeping frontend requests consistent with backend route names and expected request data.
- Connecting Node.js to MySQL with the correct host, username, password, and database name.
- Validating user input and handling missing or incorrectly formatted vehicle details.
- Displaying useful error messages when the server or database is unavailable.
- Maintaining a clear and usable interface across screen sizes.

Limitations

- The usefulness of listings depends on the accuracy and completeness of seller-provided information.
- Vehicle condition, ownership, documents, and price are not independently verified by the website.
- Online payments and advanced administration features are not included in the current version.
- The system requires a running backend and accessible MySQL database for database-backed features.

10. Conclusion:

The Second-Hand Vehicle Selling Website demonstrates how frontend technologies, a server-side application, and a relational database can be combined to support an online vehicle marketplace. It is designed to organize vehicle listings and help buyers explore second-hand cars, bikes, and trucks more conveniently.

The project combines HTML, CSS, and JavaScript for the frontend, Node.js through server.js for backend request handling, and MySQL for data storage. This separation makes it easier to maintain the interface, application logic, and database independently.

Overall, the website provides a practical foundation for an online second-hand vehicle marketplace.

11. Future Scope:

- Role-based access and stronger authentication for buyers and sellers.
- Enhanced comparison with more specifications and saved comparison lists.
- Advanced search, sorting, and location-based filtering.
- Multiple vehicle images with appropriate file validation and storage.
- An admin dashboard to review, approve, or remove listings.
- An enquiry management dashboard for sellers.
- Improved security with input validation, parameterized SQL queries, secure secret management, and rate limiting.
- Deployment to a suitable hosting environment with a managed database and backups.
- Vehicle listing status, reporting tools, and optional verification workflows.

12. References:

1. MDN Web Docs, HTML, CSS, and JavaScript: https://developer.mozilla.org/

2. Node.js Documentation: https://nodejs.org/docs/

3. Express.js Documentation (if Express is used): https://expressjs.com/

4. MySQL Documentation: https://dev.mysql.com/doc/

5. Git Documentation: https://git-scm.com/doc

6. Documentation and source files created for the Second-Hand Vehicle Selling Website project.

13. GitHub Repository Link:

The complete source code of the Second-Hand Vehicle Selling Website project is available at the following GitHub repository.

Project Source Code: https://github.com/arshad4506/second-hand-vehicle-selling-website-sap